

INTRODUCTION TO AI AND APPLICATIONS											
Course Code	25ETC15/25						CIE Marks	50			
L:T:P:S	3:0:0:0						SEE Marks	50			
Hrs / Week	03						Total Marks	100			
Credits	03						Exam Hours	03			
Course outcomes: At the end of the course, the student will be able to:											
25ETC15.1	Explain the concepts and types of artificial intelligence.										
25ETC15.2	Make use of prompt engineering techniques										
25ETC15.3	Explain the concepts of creative thinking and effective writing using prompt engineering										
25ETC15.4	Implement ethics in artificial intelligence.										
25ETC15.5	Outline recent trends in artificial intelligence.										
25ETC15.6	Identify real-world applications across different disciplines.										
Mapping of Course Outcomes to Program Outcomes and Program Specific Outcomes:											
	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11
25ETC15.1	3	2	1	-	-	-	-	-	-	-	-
25ETC15.2	2	1	-	-	-	-	-	-	-	-	-
25ETC15.3	3	3	2	1	-	-	-	-	-	-	-
25ETC15.4	2	2	-	-	-	-	-	-	-	-	-
25ETC15.5	3	2	2	-	-	-	-	-	-	-	-
25ETC15.6	2	1	-	-	-	-	-	-	-	-	-
MODULE-1 INTRODUCTION TO ARTIFICIAL INTELLIGENCE 25ETC15.1 8 Hours Introduction to Artificial Intelligence: Working of Artificial Intelligence, Advantages and Disadvantages of Artificial Intelligence, Evolution of Artificial Intelligence, Types of Artificial Intelligence, Weak AI, Strong AI, Reactive Machines, Theory of Mind, Self-Awareness, Is Artificial Intelligence Same as Augmented Intelligence and Cognitive Computing Machine Intelligence: Defining Intelligence, Components of Intelligence, Differences Between Human and Machine Intelligence, Knowledge Representation and its types, Knowledge Representation and its types											
MODULE-2 INTRODUCTION TO PROMPT ENGINEERING 25ETC15.2 8 Hours Introduction to Prompt Engineering: Introduction to Prompt Engineering, The Evolution of Prompt Engineering, Types of Prompts, How Does Prompt Engineering Work, Comprehending Prompt Engineering's Function in Communication, The Advantages of Prompt Engineering, The Future of LLM Communication Prompts for Creative Thinking and Effective Writing: Unlocking Imagination and Innovation, Brainstorming Prompts for Problem Solving, Inspiring Prompts for Artistic Expression, Igniting the Writing Process with Prompts, Prompts for Overcoming Writer's Block, Prompts for Developing Narrative Skills											
MODULE-3 PROMPT ENGINEERING TECHNIQUES 25ETC15.3 8 Hours Prompt Engineering techniques using ChatGPT and GEMINI AI and META AI: Introduction to Prompt Engineering Techniques, Instructions Prompt Technique, Zero, One, and Few Shot Prompting, Self-Consistency Prompt, Prompt Gallery Introduction to ChatGPT: Introduction to ChatGPT, A Concise History: From GPT-1 to GPT-4, How to Install and Configure ChatGPT											
MODULE-4 ETHICS AND TRENDS IN AI 25ETC15.4 8 Hours 25ETC15.5 Ethics and Trends in AI: Ethical Concerns like Privacy, sustainability, reliability and Bias, AI as a Service (AIaaS), Recent trends in AI, Expert System, Internet of Things, Artificial Intelligence of Things (AIoT)											
MODULE-5 INDUSTRIAL APPLICATIONS OF AI 25ETC15.6 8 Hours Industrial Applications of AI: Application of AI in Healthcare, Application of AI in Finance, Application of AI in Retail, Application of AI in Agriculture, Application of AI in Education, Application of AI in Transportation, AI in Experimentation and Multi-disciplinary research, Robotics-an											

Application of AI, Drones Using AI**CIE Assessment Pattern (50 Marks – Theory) –**

RBT Levels		Marks Distribution		
		Test (s)	Qualitative Assessment (s)	MCQ's
		25	15	10
L1	Remember	5	-	-
L2	Understand	5	-	-
L3	Apply	5	5	5
L4	Analyze	5	5	5
L5	Evaluate	5	5	-
L6	Create	-	-	-

SEE Assessment Pattern (50 Marks – Theory)

RBT Levels		Exam Marks Distribution (50)
L1	Remember	10
L2	Understand	10
L3	Apply	10
L4	Analyze	10
L5	Evaluate	10
L6	Create	--

Suggested Learning Resources:**Text Books:**

1. Reema Thareja, Artificial Intelligence: Beyond Classical AI, Pearson Education, 2023.
2. Ajantha Devi Vairamani and Anand Nayyar, Prompt Engineering: Empowering Communication, 1st Edition, CRC Press, Taylor & Francis Group, 2024. (DOI: <https://doi.org/10.1201/9781032692319>).
3. Saptarsi Goswami, Amit Kumar Das and Amlan Chakrabarti, “AI for Everyone – A Beginner’s Handbook for Artificial Intelligence”, Pearson, 2024.

Reference Books

1. Stuart Russell and Peter Norvig, *Artificial Intelligence: A Modern Approach* (4th Edition), Pearson Education, 2023.
2. Elaine Rich, Kevin Knight, and Shivashankar B. Nair, *Artificial Intelligence*, McGraw Hill Education.
3. Tom Taulli, *Prompt Engineering for Generative AI: ChatGPT, LLMs, and Beyond*, Apress, Springer Nature.
4. Nilakshi Jain, *Artificial Intelligence: Making A System Intelligent*, First Edition, Wiley.

Web links and Video Lectures (e-Resources):

- Elements of AI – <https://www.elementsofai.com>
- CS50’s Introduction to Artificial Intelligence with Python – Harvard <https://cs50.harvard.edu/ai/>
- Google Machine Learning Crash Course – <https://developers.google.com/machine-learning/crash-course>
- Learn Prompting (Open-Source Guide) –
- Google AI – Learn with Google AI <https://ai.google/education/>
- Coursera – Machine Learning by Andrew Ng (Stanford University) <https://www.coursera.org/learn/machine-learning>
- OpenAI Prompt Engineering Guide (for ChatGPT) <https://platform.openai.com/docs/guides/gpt-best-practices>
- Prompt Engineering for Developers – DeepLearning.AI + OpenAI <https://www.deeplearning.ai/short-courses/chatgpt-prompt-engineering-for-developers/>
- Ethics in AI – Google Responsible AI Practices <https://ai.google/responsibilities/responsible-ai-practices/>
- Google Teachable Machine (Train AI models visually without code) <https://teachablemachine.withgoogle.com>